

Developing the game Gilded Cage

Tarek Alfutih*

alfutih@chalmers.se

Chalmers University of Technology

Gothenburg, Sweden

Ida Altenstedt*

idaalte@chalmers.se

Chalmers University of Technology

Gothenburg, Sweden

Egil Guting*

guting@chalmers.se

Chalmers University of Technology

Gothenburg, Sweden

Tai-Ming Huang*

taiming@chalmers.se

Chalmers University of Technology

Gothenburg, Sweden

Adam Wikström*

adamwi@chalmers.se

Chalmers University of Technology

Gothenburg, Sweden

Rebecka Willig*

willigr@chalmers.se

Chalmers University of Technology

Gothenburg, Sweden



ABSTRACT

Gilded Cage is a new and unique take on a game within the genre *Cosy Dystopia*. It blends gameplay elements from different games to create a short gameplay experience that is intended to inspire both thoughts and conflicting feelings of calm and concern. *Gilded Cage* was created entirely from scratch using the game engine *Godot* with all assets and elements visible in the game made by the developers for the game.

The game was developed over the course of eight weeks using a mixture of different design and development practices. These include the triple diamond method, agile development, and MoSCoW. These methods, alongside two sessions to test the game and its mechanics, were used to quality control the game. This allowed for feedback on the game and helped with discovering multiple bugs. Due to this it was possible to finish *Gilded Cage* as a whole and complete experience within the allotted time frame.

KEYWORDS

game development, Godot, Aseprite, cosy-games genre, dystopian-games genre

1 INTRODUCTION

In this project the team developed a short game called *Gilded Cage*. The aim was to explore the theme cosy dystopia. Cosy dystopia is a combination of two genres, the cosy genre and the dystopian genre. Games with a cosy genre emphasise non-violence and relaxation. They usually revolve around calming or nurturing activities. While

the dystopian genre focuses on a game's theme or the setting it is in. Which means that games with both these genres are a mix between comfort and unease.

Gilded Cage might seem like a simple and relaxing on the surface but it carries a deeper meaning that players discover as they play. The aim was also to see how the cosy-dystopian feeling could be created through illustration, design, sound and storytelling. Game design can be a powerful way for opening the doors on emotions and ideas (Isbister, 2016). By combining comfortable gameplay mechanics with a hidden sense of sadness and hopelessness, *Gilded Cage* invited players to question what lies behind the relaxation and comfort they see.

The main character is a small fairy living inside a glass terrarium which was called the *Cage*. Where everything looks green and comforting, contrasting the sad and hopeless reality of being caged. The project allowed the team to practice collaboration, creative design methods, and also technical development. Multiple design and development methods have been used throughout the creation of *Gilded Cage*. Including design frameworks such as the triple diamond (Marin-Garcia et al., 2020), MoSCoW prioritisation (Kostev, 2023), and agile methods (Abrahamsson et al., 2017) such as the scrum framework, sprints and vertical slices (Ratner & Harvey, 2011).

2 RELATED WORK

Gilded Cage was inspired by a number of external sources, both games and observations from real life. However, it was primarily inspired by the award winning game *Blackwell Fishing* by Dofhouse,

*All authors contributed equally to this project.

2022, see Figure 1. *Blackwell Fishing* was used as an inspiration for the short game loop implemented in *Gilded Cage*. This was to make it possible to develop the game fully and make the experience complete with a start, middle, and end. *Blackwell Fishing* also provided the main inspiration for how dystopia was to be implemented in the game. The messages that *Blackwell Fishing*'s ending tells the player made the idea of a subtle but impactful dystopia in *Gilded Cage* attractive. This was the reason for why, if the player does not pay attention, they may entirely miss how the dystopia was incorporated into *Gilded Cage*.



Figure 1: *Blackwell Fishing*

Another piece of media that inspired a large part of *Gilded Cage* was the game *Stardew Valley* by ConcernedApe, 2016, see figure 2. *Stardew Valley* was a large inspiration for both the gameplay loop and art style of *Gilded Cage*. Since the majority of the game was intended to be a short and repeatable gameplay loop that made the player comfortable and relaxed, farming mechanics were quickly chosen for this. How they ended up working was inspired by *Stardew Valley* as that game is famous for its cosy, repeatable, and fun gameplay.



Figure 2: Image of *Stardew Valley* from the game's website

As mentioned above, the graphics of *Gilded Cage* was partially inspired by *Stardew Valley*, however, this was not the only game that inspired this aspect of the *Gilded Cage*. The art style of the game was largely inspired by three different games, *Stardew Valley*, *CrossCode* by Radical Fish Games, 2018, and the early *Pokémon Mystery Dungeon (PMD)* games by Spike Chunsoft, 2007-2009, more specifically *Time*, *Darkness*, and *Sky*. Using these games as inspiration it was possible to create an environment that, while being

2D, did not look flat and vertical through clever use of perspective and shadows. The character's animations were further inspired by how both *Stardew Valley* and *PMD* implemented their animations depending on the direction the player is facing.

Another game that inspired the dystopian aspect of *Gilded Cage* is *Outer Wilds* by Mobius Digital Games, 2019, a game that rewards curiosity and invites reflection. Although it is not a dystopia, it uses a loop as a narrative device, where it can feel like no matter what you do, when you wake up you are back where you started. Furthermore, just like in *Gilded Cage*, the ending is hidden from the player, and they have to explore and understand the world in order to find it, and once they do it might not be what they originally thought.

The music of *Gilded Cage* was primarily inspired by games in the cosy genre, such as *Stardew Valley*, using bright and calm melodies that make the player feel a sense of peace or tranquillity. This is contrasted by the music played during the end credits, which is a slow and sad version of the main theme. The credits theme is, in fact, created using the same melody as the main theme, but shifted to a minor mode – an idea that was inspired by *Outer Wilds* which does something similar to its main theme during another piece of music in the game.

Outside of games there were other things that also inspired *Gilded Cage*. The cage itself was inspired by enclosed terrariums since the world was supposed to create the feeling that the fairy living in a self-sustaining ecosystem. The fairy's house and the bench inside of the cage was inspired by real world doll houses and a plastic toy one of the authors had as a child. Due to this they were designed to match that aesthetic.

3 DEVELOPMENT PROCESS

The triple diamond design sprint methodology was used for this development process. It includes six phases – understand, define, sketch, decide, prototype, and validate. The triple diamond methodology is also showcased in Figure 3. The focus was mainly put on understanding the theme, defining the game concept, sketching ideas, and deciding which idea to implement.

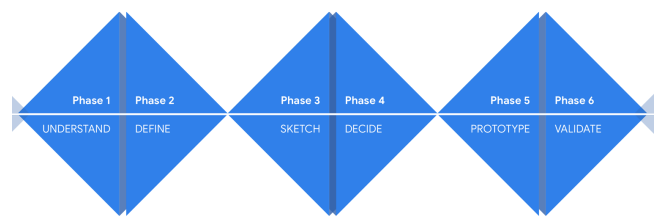


Figure 3: Image of the triple diamond design process

Design

The process started with the ideation phase, where the main task was to understand the concept of cosy dystopia. Three different ideas were defined, and developed to be explained during a pitch presentation. The first idea, *Animal Island*, had the player playing

as the only person on a lonely island, working to bring life back by caring for the plants and animals. It was meant to have a cosy experience set in a ruined world. The second idea, *Gilded Cage*, was about a fairy trapped in a small terrarium on a window ledge. With the twist being that she had lost hope of being rescued and decided to take her own life. The last idea, *Trashcoon*, was meant to take place in a world full of garbage where humans no longer exist. The gameplay would revolve around the player building a new home by recycling waste. Even though the ideas had dystopian settings, the gameplay was meant to feel cosy.

After discussing as a team, it was decided to move forward with the *Gilded Cage* concept. Many sketches were created to explore its elements until the concept became clearer. The decision phase was conducted using the MoSCoW Method (Kostev, 2023) to limit the scope of the project. After this, several aspects that were required to implement the game were identified. These included a story and scenario; drawings and animations; sound effects and music; and finally, the implementation of the gameplay and all these parts. To cover all these aspects the developer team split into smaller groups that each focused on developing different aspects of the game. The story and scenario was finalised during the decision step.

Implementation

When it came to executing the prototype and validate phases, they were handled through agile development, specifically using the Scrum framework. Which, for this project, meant that weekly sprints were utilised to iterate over the third diamond. A *Trello* (Atlassian, n.d.) board was set up to support the process, ensuring that all members were up to date, had clearly defined tasks, and that several people did not work on the same task. The group worked together in sprints – at the beginning of each week, tasks for that week were chosen and each developer selected a set of tasks to take on. At the end of the week, the progress was reviewed and all completed tasks were marked as such, and new tasks would be introduced, in accordance with the MoSCoW. This resulted in a structured development process that made sure that all members always had a task to work on and that said task would be finished. As such, the work was done using the Scrum framework for agile development and the weekly sprints made good use of the vertical slicing (Ratner & Harvey, 2011) method, allowing the group to fully fledge out each feature efficiently. Once the *Trello* board had been set up, the group split up into smaller teams to start working on the different parts of the projects.

GitHub (GitHub, Inc., n.d.) was used for version control, and a repository was set up and used throughout the project. This let the art and sound teams share their assets with the programmers, and let the programmers work simultaneously on the project.

The art team used *Aseprite* (“Aseprite”, n.d.) to create most visual assets for the game. This was especially useful since *Aseprite* is focused on creating pixel art, which is what the game was decided to use. *Aseprite* was used both to create static images such as the ground and the cage, as well as animations such as the character and flowers. *Adobe Photoshop* (Knoll, 1989) was also used to create the image in the main menu of the game, see Figure 4.

The game’s sound effects were recorded by the sound team using *Adobe Audition* (Adobe Inc., 2003) and *Audacity* (“Audacity ®”, n.d.),



Figure 4: *Gilded Cage* main menu

and later edited using *Audacity*. *Noteflight* (“Noteflight”, n.d.) and *Cakewalk Sonar* (“Cakewalk Sonar”, n.d.) were used to compose and produce the music, see Figure 5. To produce the music in *Cakewalk Sonar*, samples of three instruments were used, one was built-in in the software and two were free VST plugins – *Etherealwinds Harp II* (“Etherealwinds Harp II: CE - Versilian Studios”, n.d.) and *Sonatina Flute* (James, n.d.). These sampled instruments were the only aspects of the aesthetics that were not created by the group themselves.

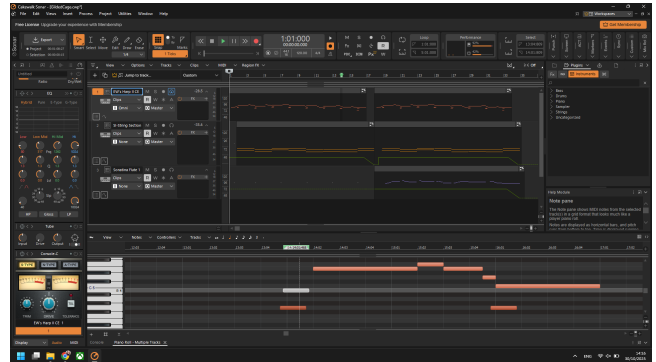


Figure 5: Production of the main theme in *Cakewalk Sonar*

Godot (“Godot Engine”, n.d.) was the chosen game engine for this project. This was because out of all game engines that some group members had experience with, *Godot* was argued to be the quickest and easiest to learn for those who lacked prior experience. Using *Godot*, scenes and scripts were created by the programming team, and then linked together using these scripts as well as signals. These scenes include, but are not limited to, each piece of decor, interactable object and sound effect, as well as the playable character, main menu and credit sequence, which all had their own scene and most of which also had a script – code that regulated what these scenes could do and when.

4 EVALUATION

The game has been evaluated mostly through play-testing. Be it organised with outside users as the players, or simply the developers themselves testing components of the game as they make

them. There were two play-testing sessions during the development, mostly focused on usability testing, and one larger evaluation after the game was considered finished.

The evaluations were conducted through letting the players explore the game on their own with a developer ready to answer questions or give hints where needed. The first play-test had a questionnaire which the players were requested to answer after testing the game. The questionnaire was made as an attempt to gain quantitative data for specific aspects of the game while also having fields for more open responses. When going over the feedback it was quickly determined that a more qualitative response would be preferable, since the more open-ended questions got more expressive and varied responses. For some of these responses it would have been useful to explore them further while other feedback was simply remarking on aspects that were already scheduled for implementation.

Therefore the second play-test was conducted fairly similarly to the first with the difference being that the questionnaire was replaced with an informal interview between one of the developers and the play-tester, using pen and paper to take notes. This allowed for more qualitative data gathering where useful feedback could be expanded upon. The addition of writing instruments at the testing area also resulted in general notes being taken during the observation of the play-tests. These notes included anything from remarking on when hitboxes were not obvious; and places where extra hints were needed. To when an animation could be further tweaked, or where an asset was missing.

During these two play-testing sessions the poison plant – the catalyst for accessing the ending – was deliberately removed as to not reveal how the game would end and allow for genuine reactions during the final evaluation. Something which resulted in there being a lack of testing for this part of the game, which was reflected in the final evaluation.

Exhibition

The final evaluation took place during the game exhibition and resulted in many hours worth of valuable feedback. The play-testing was further refined by splitting the tasks of note-taker and play-testing overseer. Which meant that one developer could focus on observing the players and how they interacted with the game while the other observed the players themselves and focused on helping them as required.

Below is a list of some of the information gained during the final evaluation:

- The main gameplay loop could be done through blindly following the thought bubbles instead of thinking on their own and taking in the world.
- The thought bubble used when showing that a plant did not require any more water was misleading, leading to many players attempting to refill their water bucket instead of exploring the map and looking for more flowers to water. (The crossed out water droplet in Figure 6 is the confusing icon referenced here.)
- Some of the players had difficulty realising when they were meant to put down the watering can to continue forward.
- Confusion for what could be interacted with since the interact button also brought up a thought bubble, resulting in many trying

to interact with non-interactable sprites or missing interactable items.

- A few players tried to find meaning or symbolism in the colours used throughout the game.
- The players felt that interactions in the game were limited. They lacked actions outside of those that contributed to progressing forward, they wanted to interact with objects such as shelves or perform general, non-essential actions.
- Many players didn't realise that the fairy took suicide instead of just a nap, something that was not helped by the screen going dark too fast for them.
- Many players entire missed that there were a third flower before told to look for it, if this was because they thought they lacked water or because the flower was hard to find is unknown.
- The secret ending was hard to find for many players.
- The exploration aspect was well liked.
- The graphics and character design were well liked.
- The players liked the music and how both it and silence was utilised in the game.
- The gameplay mechanics were fun.
- Players that understood the story also remarked that they liked said story.
- The game had a comfortable and cosy feel to it.

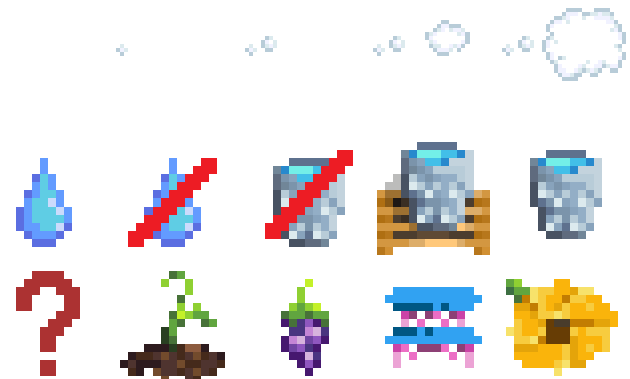


Figure 6: The thought bubble and its icons

5 RESULT & DISCUSSION

The finalised game has a tile-based play area with greenery including but not limited to, bushes, flowers, grass and dirt tiles. Some of these flowers have multiple sprites showing the flower in different stages of growth. A watering can, in the shape of a thimble, can be picked up which allows the player to water the flowers. This also plays an animation showing the flowers growing. The game allows the player character to interact with objects and there are animations as well as sound effects for these interactions. For example, interacting with the watering can causes an animation of the player character picking it up, as well as a sound, to play. The player character also uses three different sets of animations. Each of these representing different states of the movement; moving, standing still and a state when transitioning between the moving and still state. Two music tracks have been implemented, one playing at the

start of the game and one playing in the credits. A particle system was also made which creates small fading particles.

A repeatable gameplay-loop exists where the fairy can water the flowers each day, eat a fruit from one of them and then sit on a bench to pass the day. This zooms the camera out, allowing the player to view the cage. It is also possible to water another partly obscured flower and eat its fruit. This kills the player and shows the credits. A guiding system was also created. When the player presses the button needed for interacting and no interaction is possible it animates a thought bubble showing them what should be done next for the gameplay loop. The sprites for the thought bubble and its content is shown in Figure 6. Next to the obscured flower this would animate a question mark if no other interaction was possible.

The final product fulfilled the requirements that was intended when developing it. The intention from the beginning was to create a short game that was completely finished and had an ending players could reach. Some things proposed as ‘should have’ or ‘could have’ from the MoSCoW were not implemented in the game but all ‘must haves’ have been created. Given more time the most critical changes to make would be to fix problems players encountered during the exhibition.

Feedback

While players generally were able to understand what to do, there were multiple times where interactions were missed because players were too far away from objects to trigger the interactions. At other times players felt encouraged to continue the gameplay loop, expecting something to change after enough iterations had been made. While this reinforces the aesthetic feeling of a monotone loop it also stopped players from exploring and therefore from realising that the obscured flower could be watered. Encouraging players to explore more by creating more things to interact with outside the gameplay loop could be a possible way to avoid this problem. Another problem was the fade to black which happened when the player started the ending. While this was meant to simulate the feeling of slowly dying, it also directly impacted visibility during the scene making the player both not understand that the player character was dying and what the fade to black exactly represented. The lack of context and lore also lead to players providing their own interpretations of what the story and its themes were. While this process was a gratifying part of watching others play the game, making the ending more visible for players would have allowed them to interact with a more concrete thematic structure and possibly create a more satisfying ending for the players. Players also suggested providing insight about the fairy’s mental state through different means. This could provide avenues for expressing dystopia and help discouraging the gameplay loop by showing perhaps showing a disinterest from the fairy. It would also fit well with the already implemented thought bubbles since they could show these thoughts to the player. However, it might also stop players from interpreting these elements by themselves and remove some shock value from the ending. Furthermore, the comfy aesthetic could be impacted by this. By inserting more blatant representations of sadness or dystopia into the game it could disrupt the relaxing feelings that are built through the aesthetic gameplay. As such, if more hints

towards the story and dystopia were to be implemented it would be important to do so carefully to not damage the cosy aesthetic.

6 FUTURE WORK

If game development was continued, there were three categories of possible avenues that could be explored; polishing the game as a complete product, extending it to include more content, or adding content for post-game. Below are potential tasks for these paths.

Polish and Core Features

Changing or expanding upon already existing features to improve player experience.

- **Audio (Ambience):** Implement subtle "Fairy bells" audio cues occasionally for when the fairy moves.
- **Audio (Interaction):** Add a soft sound of something that plings to indicate that an object can now be interacted with.
- **Audio (Dynamic Music):** Dip the music volume during the night sequence.
- **Save Functionality:** Implement a feature to save and load game progress.
- **Accessibility Features:** Add options such as text resizing, colour-blind modes, or re-mappable controls.
- **Technical (Performance):** Optimise game performance and loading times.
- **Localisation:** Translate the game text into other languages to reach a wider audience.

Content and Narrative Expansion

Features that would add to the gameplay and/or provide more story to the game.

- **Narrative Expansion (Butterfly):** Develop and implement multiple or different endings based on player actions.
- **Passing person:** Showing the shadow of a person passing by the cage at times.
- **Gameplay Expansion (Magic Flower):** Introduce more kinds of flowers, each possibly having unique effects or narrative roles.
- **Lore and Tracking (Bookshelf):** Add a bookshelf containing a diary, with new entries appearing as days pass in the game.
- **Day count:** Have tally marks on the inside wall of the house. These can then increase by one each day played.
- **Window:** Have a window showing the outside environment, making it more obvious the fairy is inside a house
- **Seasonal change:** Allowing changes in the seasons that are viewable through the window if implemented.

Post-Completion Features

Features that unlock after a player has finished the game.

- **Post-Game Audio (Credits Music):** Pressing credits plays the sad music after completion.
- **Post-Game Visuals (Background Change):** The background should change to see the cage after completion.

7 CONCLUSION

This project report describes the journey of developing the game *Gilded Cage*, from a theme and concept through various tools all the way to a finished product.

Gilded Cage interprets the theme of cosy dystopia by letting the aesthetics and gameplay be cosy and the dystopian aspect come from the narrative. The game took inspiration from games in the cosy genre such as *Stardew Valley*, especially when it comes to the aesthetics and gameplay, and its short game loop was primarily inspired by *Blackwell Fishing*.

Through agile development and weekly sprints, the project was able to be completed in eight weeks. By using various development software tools such as *Aseprite* and the game engine *Godot*, the development team created an immersive experience where the player gets to live the life of a fairy trapped in a cage with no escape.

REFERENCES

- Abrahamsson, P., Salo, O., Ronkainen, J., & Warsta, J. (2017). Agile software development methods: Review and analysis. *arXiv preprint arXiv:1709.08439*.
- Adobe Inc. (2003). Adobe audition [Accessed: 2025-10-30]. <https://www.adobe.com/products/audition.html>
- Aseprite [Accessed 23-10-2025]. (n.d.). <https://store.steampowered.com/app/431730/Aseprite/>
- Atlassian. (n.d.). Trello [Accessed: 2025-10-30]. <https://trello.com>
- Audacity® [Accessed 30-10-2025]. (n.d.). <https://www.audacityteam.org/>
- Cakewalk Sonar [Accessed 24-10-2025]. (n.d.). <https://www.cakewalk.com/sonar/>
- ConcernedApe. (2016). *Stardew Valley* [Accessed 30-10-2025]. <https://www.stardewvalley.net>
- Dofhouse. (2022). *Blackwell Fishing* [Accessed 24-10-2025]. <https://dofhouse.itch.io/blackwell-fishing>
- Etherealwinds Harp II: CE - Versilian Studios [Accessed 30-10-2025]. (n.d.). <https://versilian-studios.com/etherealwinds-harp/>
- GitHub, Inc. (n.d.). Github [Accessed: 2025-10-30]. <https://github.com>
- Godot Engine [Accessed 23-10-2025]. (n.d.). https://store.steampowered.com/app/404790/Godot_Engine/
- Isbister, K. (2016). *How games move us: Emotion by design*. Mit Press.
- James, B. (n.d.). PLUGINS4FREE [Accessed 30-10-2025]. <https://plugins4free.com/plugin/2312/>
- Knoll, J. (1989). Adobe photoshop [Accessed: 2025-10-30]. <https://www.adobe.com/products/photoshop.html>
- Kostev, R. S. (2023). Challenges and problems of the moscow method application in erp system implementation. *2023 International Scientific Conference on Computer Science (COMSCI)*, 1–4.
- Marin-Garcia, J. A., Garcia-Sabater, J. J., Garcia-Sabater, J. P., & Maheut, J. (2020). Protocol: Triple diamond method for problem solving and design thinking. rubric validation. *WPOM-Working Papers on Operations Management*, 11(2), 49–68.
- Mobius Digital Games. (2019). *Outer Wilds* [Accessed 30-10-2025]. https://store.steampowered.com/app/753640/Outer_Wilds/
- Noteflight [Accessed 24-10-2025]. (n.d.). <https://www.noteflight.com/>
- Radical Fish Games. (2018). *CrossCode* [[Accessed 30-10-2025]].
- Ratner, I. M., & Harvey, J. (2011). Vertical slicing: Smaller is better. *2011 Agile Conference*, 240–245. <https://doi.org/10.1109/AGILE.2011.46>
- Spike Chunsoft. (2007-2009). *Pokémon Mystery Dungeon* [[Accessed 30-10-2025]]. https://bulbapedia.bulbagarden.net/wiki/Pok%C3%A9mon_Mystery_Dungeon_series